Fine-grained Gesture Recognition Using WiFi

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***Abstract*—Gesture recognition has drawn increasingly attention in human-computer interaction (HCI) and can support a variety of emerging applications such as smart home and mobile gaming. Traditional approaches usually involve wearable sensors and specialized hardware installations. This paper presents fine-grained finger gesture recognition by using a single commodity WiFi device without requiring user to wear any sensor. Our low-cost system, WiFinger, takes advantages of the detailed channel state information (CSI) available from commodity WiFi devices and the prevalence of WiFi infrastructure. It senses and identifies subtle movements of finger gestures by examining the unique patterns exhibited in the detailed CSI. We devise environmental noise removal mechanism to mitigate the effect of signal dynamic due to the environment changes. We also design algorithm to capture the intrinsic gesture behavior to deal with individual diversity and gesture inconsistency. Our experimental evaluation in a home environment demonstrates that our system can achieve over 93% recognition accuracy and is robust to both environment changes and individual diversity.**

I. Introduction

Gesture recognition is gaining increasing importance in human-computer interaction (HCI). Gesture based interaction serves as a convenient and natural means for users to interact with computers. Gesture made with fingers is particularly crucial to interact with mobile and wearable devices and to perform finger control in smart home and mobile gaming. Google's Soli radar chip [1], for example, is recently developed for the wearables to recognize finger gestures.

Existing gesture recognition solutions primarily rely on dedicated sensors worn by the subject [2] or cameras installed in the environment (e.g., leap motion) [3]. These systems either require significant deployment effort or incur non-negligible cost. Recently, radio frequency (RF) based gesture recognition [4], [5], [6] have drawn considerable attention as they don't require users to wear any physical sensors. These systems however only provide coarse-grained gesture recognition such as body activities [6], [4] or hand movements [5].

This work demonstrates that the commodity WiFi can be exploited for fine-grained finger gesture recognition which is both easily deployable and low-cost. It exploits two novel points in WiFi technology. First, the detailed physical layer measurement of CSI is internally tracked by IEEE 802.11 MIMO and is readily available in commodity WiFi devices. For example, the 802.11a/g/n/ac employs OFDM technology which partitions the relatively wideband 20MHz channel into 52 subcarriers and provides detailed CSI of each subcarrier. The detailed CSI is able to detect the minute environment changes that alter signal propagation and multipath. It is thus capable of capturing the subtle movements of fingers to provide fine-grained gesture recognition. Leveraging detailed CSI to recognize gestures doesn't require users to wear any sensors. Second, the prevalence of WiFi infrastructure enables the proposed system to reuse existing WiFi devices and networks without requiring dedicated or specialized hardware. The system could reuse existing WiFi signals, for example, the WiFi beacons, to perform gesture recognition.

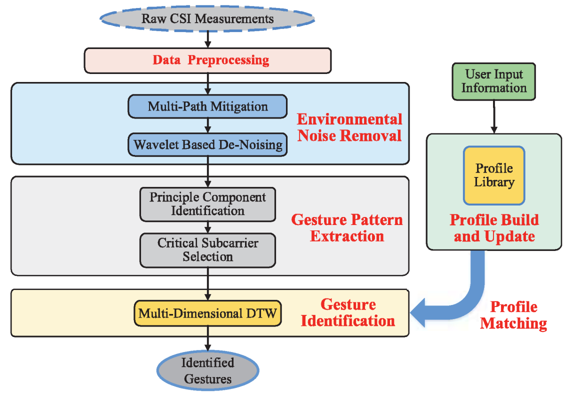


Fig. 1: Overview of system flow.

In particular, our system, WiFinger, uses only a single commodity WiFi device with its connected AP to recognize finger gestures by examining the unique patterns exhibited in the detailed CSI. Accurately discerning the gestures is challenging because the multipath, shadowing, and fading components of signal could be dynamic due to the environment changes. For example, people walking around or the moved furniture could change the multipath and affect the signal propagation. Such changes could also be sensed by the detailed CSI and may distort the CSI pattern of the finger gesture. We thus propose an environmental noise removal mechanism to filter out the environmental noise while retaining the CSI pattern only resulted from the finger gesture.

Moreover, there exists individual diversity of each user such as finger length, movement speed and gesture consistency. We introduce gesture pattern extraction algorithms to identify the principle components of the CSI pattern and to choose critical subcarriers that are sensitive to the finger gesture for accurate gesture recognition. We experimentally evaluate it with eight typical figure gestures in a home environment. Results show that our system achieves overall accuracy over 93% and is robust to both environment changes and individual diversity.

II. System Design

The basic idea of our system is to sense and identify subtle movements of finger gesture by examining the unique patterns exhibited in the detailed CSI. As each finger gesture has its unique CSI pattern, the gesture recognition could be done by matching the CSI pattern against the gesture profiles. As shown in Figure 1, the system takes as input time-series CSI measurements extracted from one single WiFi device with its connected AP. It can reuse existing WiFi traffic for extracting the CSI. The measured CSI is then preprocessed to remove out-of-band noise via a butterworth filter and to normalize the CSI traces to the same scale.

The core of our system consists of the *Environmental Noise Removal* and the *Gesture Pattern Extraction*. It first employs *Multipath Mitigation* to mitigate the interference stemmed from the environment changes such as moved furniture and/or people moving around. The environment changes will reflect the wireless signals and creates additional multipath, which distort the CSI pattern of a finger gesture. In particular, the signal reflection via multipath usually has longer propagation delays before arriving the receiver. By transferring the frequency domain CSI into time-domain power delay profile, we remove the signal components that have longer delay to mitigate the effect of the changed environment.

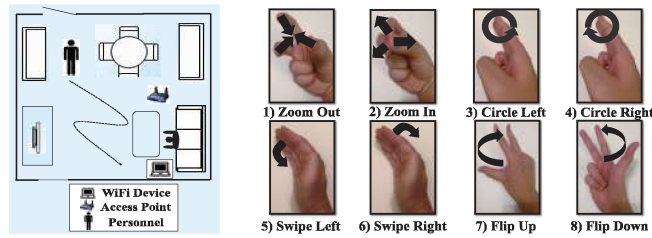
Our system then utilizes *Wavelet Based Denoising* to further remove the noises presented in the CSI measurements. These noises could come from various sources such as the nearby electric devices. It is based on the Discrete Wavelet Transform (DWT) which decomposes signals into approximation and detail coefficients. While the approximation coefficients describe the shape/trend of the signal, the detail coefficients capture both high frequency noise and the fine details of the CSI pattern. To remove the high frequency noise components while retaining sufficient details for differentiating similar gestures, a dynamic thresholding is applied to the detail coefficients to remove the noise components.

Next, our system performs gesture pattern extraction by utilizing *Principle Component Identification* and *Critical Subcarrier Selection*. The system captures the intrinsic user gesture behaviors by identifying the CSI components which remain stable within the user's finger gestures. Particularly, we compare the CSI patterns of multiple instances of the same gesture and identity the components without significant distortion as the principle components. The identified principle components are invariant in the presence of extensive variations in the user's gesture, and hence resilient to individual diversity. Our system then identifies the subcarriers that have high sensitivity to the subtle figure movements for gesture recognition.

At last, WiFinger recognizes gestures by calculating the similarity between the extracted CSI pattern and the pre-constructed gesture profiles. *Muti-Dimensional Dynamic Time Warping (MD-DTW)* is used to calculate similarity by aligning CSI pattern to the gesture profiles while correcting for difference in finger movement speed. The one with the profile in the library that has the highest and sufficient similarity with the testing CSI pattern is then identified as the recognized gesture.

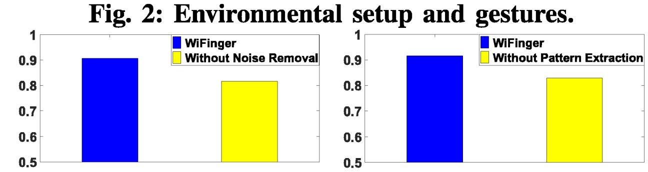
III. Performance Evaluation

We conduct experiments with a laptop (i.e., Dell E5540) connected to a commercial AP (LINKSYS E2500) in a home environment, as shown in Figure 2(a). The package transmission rate is set to 20 pkts/s. For each packet, we extract CSI for 30 subcarrier groups. The environment is about 16 ft by 13 ft with regular living room furniture setup. The participant



(a) Experimental setup (b) Gestures

Fig. 2: Environmental setup and gestures.



(a) People walking around (b) Individual diversity

Fig. 3: Overall accuracy with people walk around and individual diversity.

performs eight types of finger gesture, as shown in Figure 2(b), when sitting on the sofa. The AP and the laptop are placed at two sides of the sofa. We collect data from five participants and each performs every gesture for fifty times. We use ten instances of each gesture to construct the profile.

We observe that our system achieves an overall recognition accuracy over 93%. In particular, the swipe left/right has the overall accuracy of 95% while the others have the accuracy ranging from 91% to 94%. This is due to the relative larger finger movements involved in swipe left/right.

We also evaluate the robustness of our system to the environment changes. We introduce environment changes by asking people to walk around when the participant is doing finger gestures. We then compare the performance of our system with and without using environmental noise removal technique under the environmental changes, as shown in Figure 3(a). We find that our system, WiFinger, has the accuracy above 90% while the performance is below 80% without using environmental noise removal technique.

We further test the resilience of our system to individual diversity by applying the gesture profile built from one participant to another. We compare the performance of our system to the one without using the gesture pattern extraction method, as shown in Figure 3(b). We observe that without using gesture pattern extraction method, the accuracy degrades to around 80%. Our system provides more than 90% recognition accuracy with pattern extraction method. Above results show that our system provides accurate fine-grained finger gesture recognition and is robust to both environment changes and individual diversity.

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